

Instruction manual

Two longed Arch



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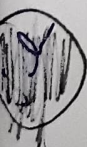
TWO HINGED ARCH APPARATUS

INSTRUCTION MANUAL

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Two Hinged Arch Apparatus

1.0 APPARATUS REQUIRED:

- Two Hinged Arch Apparatus
- Dial Gauge : 1 No.
- Magnetic Base : 1 No.
- Hangers : 8 Nos.

2.0 THEORY:

The two hinged arch is a statically indeterminate structure of the first degree. The horizontal thrust is the redundant reaction and is obtained by the use of strain energy methods.

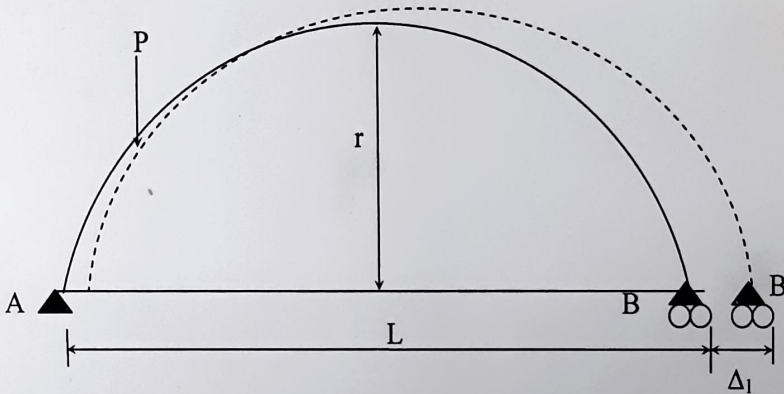


Fig (a)

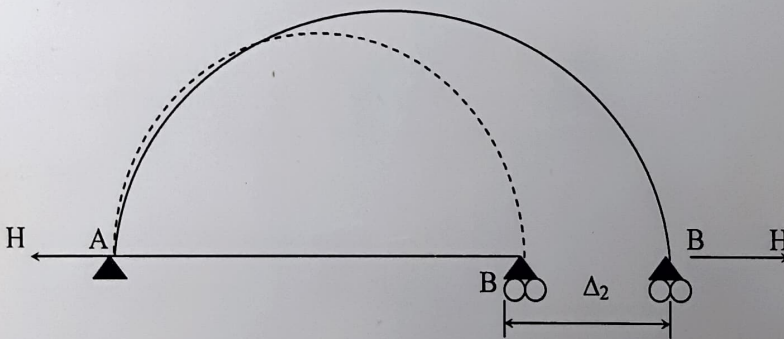


Fig. (b)

Two hinged arch is made determinate by treating it as a simply supported curved beam and horizontal thrust as a redundant reaction. The arch spreads out under external load as shown in fig. (a). This results in a horizontal displacement of support B by Δ_1 . Here, deflection due to flexure only has been considered. Since

the support conditions dictate that the final displacement at support B should be zero, horizontal reaction H should be such that displacement Δ_2 caused by H must satisfy the condition.

$$\therefore \Delta_1 + \Delta_2 = 0$$

$$\Delta_1 + f \times H = 0$$

Where, f is the displacement caused by a unit force applied in the direction of H.

$$\therefore H = -\frac{\Delta_1}{f} \quad (1)$$

Therefore, it is required to calculate the horizontal displacement in arch caused by external load as well as unit horizontal force.

The horizontal displacement in a curved member can be found by either Castiglano's second theorem or the unit load method.

$$\begin{aligned} \therefore H &= \int_A^B M \frac{\partial M}{\partial H} \frac{ds}{EI} \\ &= \int_A^B Mm \frac{ds}{EI} \\ \text{similarly } f &= \int_A^B m^2 \frac{ds}{EI} \\ \text{therefore, } H &= -\frac{\int \frac{Mm ds}{EI}}{\int \frac{m^2 ds}{EI}} \quad (2) \end{aligned}$$

Where, M = Bending moment on any point on the arch due to given loading.

m = moment on any point on the arch due to a unit horizontal force

Applied at B in the direction of H.

The expression given by Eq. (2) will become simpler provided the curve of the arch axis is parabolic and moment of inertia of curve at any section varies as $I = I_0 \sec\theta$ where θ is angle between the horizontal and tangent to the arch axis at that particular point.

I_0 = moment of inertia at the crown

I = moment of inertia at any other section = $\frac{bd^3}{12}$

m = y,

$ds = \sec\theta dx$

NOMENCLATURE:

L = cm

$$\begin{aligned}
 b &= 4 \text{ cm} - 40 \text{ mm} \\
 d &= 10 \text{ mm} \\
 \theta &= 1.2 \\
 r &= 27 \text{ cm}
 \end{aligned}$$

it may be noted that the integration is to be carried out from 0 to L then Eq. (2) will become

$$H = - \frac{\int_0^L My dx}{\int_0^L y^2 dx} \quad (3)$$

$$\text{And } \Delta_1 = \text{horizontal displacement} = \int_0^L \frac{my dx}{EI_0}$$

For a concentrated load W at the crown it is found that

$$\text{Horizontal displacement } \Delta_1 = \frac{5 WL^2 r}{48 EI_0} \quad (4)$$

Where, L is the span of the arch and r is the rise.

The horizontal movement of the roller end can be found by this method for any position of the load on the arch.

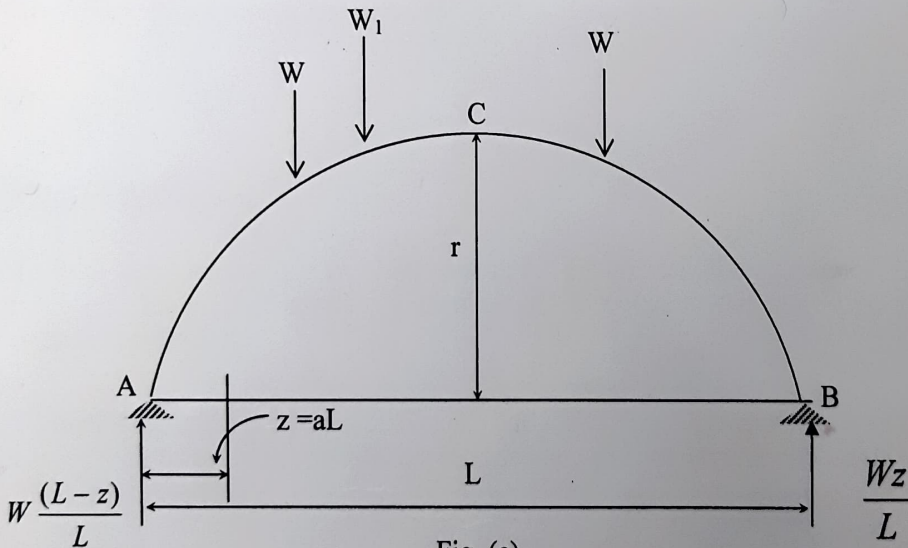


Fig. (c)

The ordinate for the influence line diagram for H at any distance $z = aL$ from L.H.S. can be obtained as follows:

$$M = \frac{Wx}{L} (L - z) \text{ for } 0 < x < z$$

$$M = \frac{Wz}{L} (L - x) \text{ for } z < x < L$$

Now H can be evaluated using Eq. (3)

$$\int_0^L My \, dx = \int_0^z \frac{Wx}{L} (L-z) 4r \left(\frac{x}{L} - \frac{x^2}{L^2} \right) dx + \int_z^L \frac{Wz(L-x)}{L} 4r \left(\frac{x}{L} - \frac{x^2}{L^2} \right) dx$$

$$= \frac{W_z(L-z)(L^2 + Lz - z^2)}{3L^2}$$

$$\int_0^L y^2 \, dx = \int_0^L \left[4r \left(\frac{x}{L} - \frac{x^2}{L^2} \right) \right]^2 dx$$

$$= \frac{8}{15} Lr^2$$

$$\therefore H = -\frac{5}{8} \frac{Wz}{rL^3} (L-z)(L^2 + Lz - z^2)$$

Now substituting $z = aL$

We have $H = -\frac{5}{8} \frac{WL}{r} (a - 2a^3 + a^4)$

Taking $W=1\text{kg}$ or increase gradually

$\therefore$ Influence line ordinate are given by

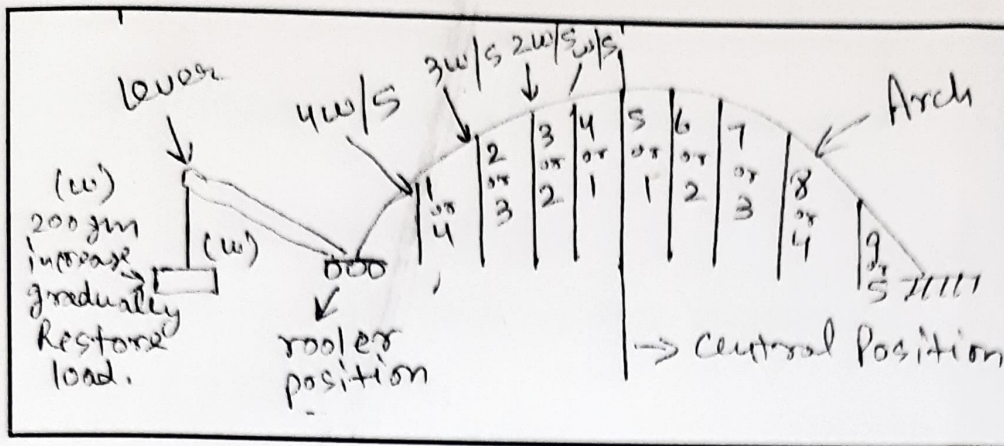
$$ILO = \frac{5}{8} \frac{L}{r} (a - 2a^3 + a^4) \quad (5)$$

3.0 OBJECTIVE:

1. To study experimentally a two hinged arch for the horizontal displacement of the roller end for a given system of loading and to compare the same with those obtained analytically.
2. To obtain the influence line ordinates for horizontal thrust experimentally in a two hinged arch by moving a load along the span and to compare the same with those obtained theoretically.

4.0 APPARATUS:

The model has a span of 100cm and rise 25cm. Both ends are hinged but one of the ends is also free to move longitudinally. A lever arrangement is fitted at this end for the application of known horizontal inward force for measuring the horizontal thrust. Along the horizontal span of the arch various points are marked at equidistant for the application of load. This being a statically indeterminate structure of the first degree. A dial gauge with magnetic base is supplied with the apparatus.



- $\frac{3w}{4}$ = First Hanger Position
- $\frac{2w}{4}$ = Second Hanger Position
- $\frac{w}{4}$ = Third Hanger Position

5.0 SUGGESTED EXPERIMENTAL WORK:

For Table 1

1. Firstly mount the apparatus to remove the play and leveling the apparatus or equipment.
2. Fix the hanger (7 Nos.) to apply the load gradually or tabulated method of the arch.
3. Fix the dial gauge to measure the movement of the roller end of the model and keep the lever out of contact.
4. Place a load of 0.5kg on the central hanger of the arch to remove any slackness and taking this as the initial position, set the reading on the dial gauge to zero.
5. Now add 1kg weights to the hanger and tabulated the horizontal movement of the roller end with increase in the load in steps of 1kg. Take the reading up to 5kg load. Dial gauge reading should be noted at the time of unloading also.
6. Plot a graph between the load and displacement (theoretical and experimental) compare. Theoretical values should be computed by Eq. (4).

For Table 2

7. Remove control load $\frac{1}{2}$ kg was applied of each contact the lever and apply 200 gm wt or more as thread of lever. Set dial to zero.
8. Now move the lever in contact with 200gm hanger on ratio 4/1 position with a 1kg load on the first hanger. Set the initial reading of the dial gauge to zero.
9. Restore the dial gauge reading to zero by adding loads to the lever hanger, say the load is w kg.
10. The experimental value of the influence line ordinate at the first hanger position shall be $\frac{3w}{4}, 2^{nd} \frac{2w}{4}, 3^{rd} \frac{w}{4}$ etc.

11. Repeat the steps 5 to 7 for all other hanger loading positions and tabulate. Plot the influence line ordinates.
12. Compare the experimental values with those obtained theoretically by using Eq. (5).

6.0 RESULTS AND DISCUSSIONS:

Compare the two results obtained theoretically and analytically.

7.0 SAMPLE DATA SHEET:

Table-1
Horizontal displacement

Sl. No.	Central load (kg)	0.0	1.0	2.0	3.0	4.0	5.0	6.0
1	Observed horizontal displacement (mm)							
2	Calculated horizontal displacement Eq.(4)							

Table-2
Influence Line Ordinate

Sl. No.	Load 5kg applied to hanger number	1	2	3	4	5	6	7
1	Weight w kg required on the lever to balance load							
2	Influence line ordinate = $3w/4$ Calculated ILO using Eq. (5)							

8.0 PRECAUTIONS:

- Apply the loads without jerk.
 - Perform the experiment away from vibration and other disturbances.
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